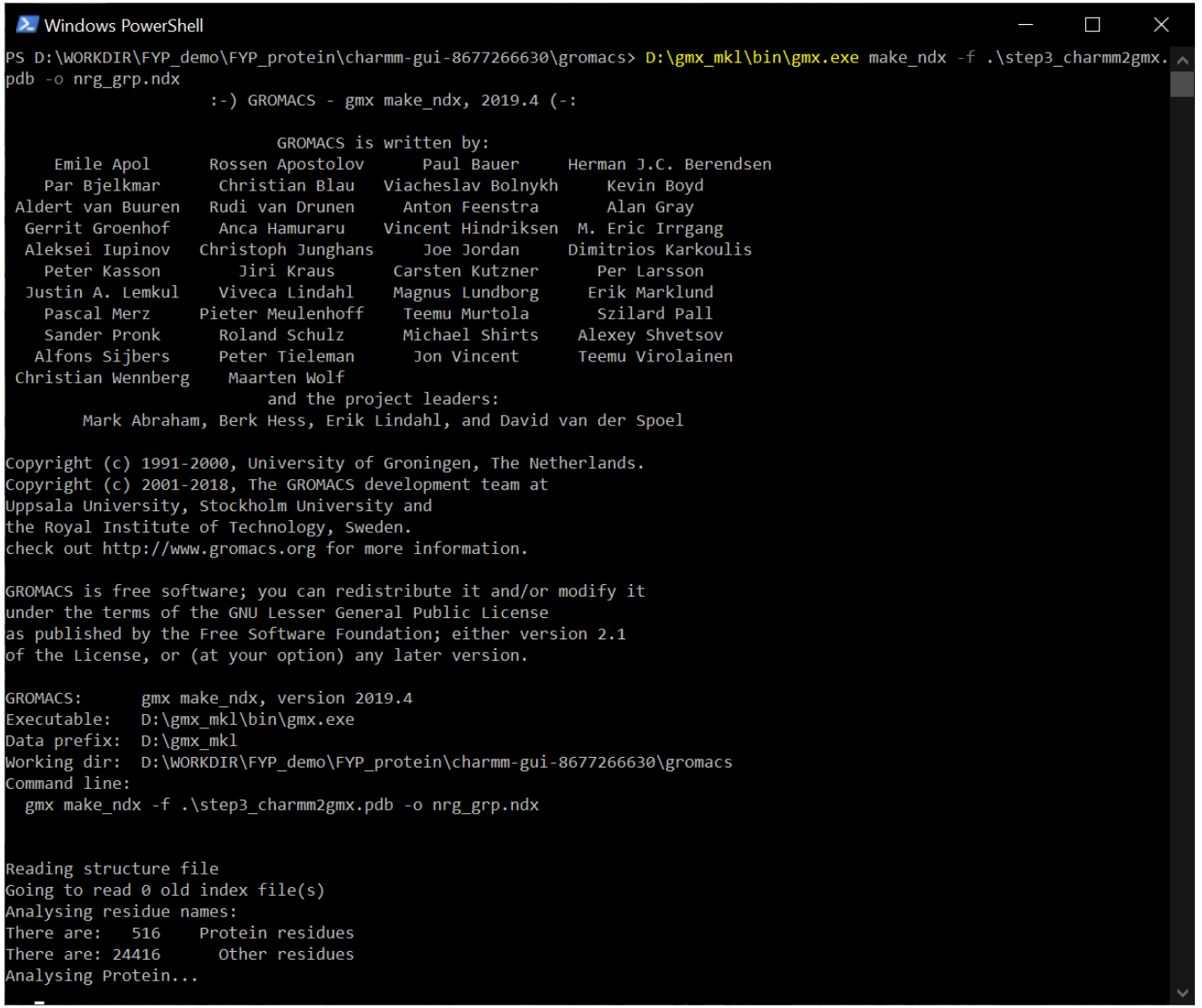
3bi. Generate index groups

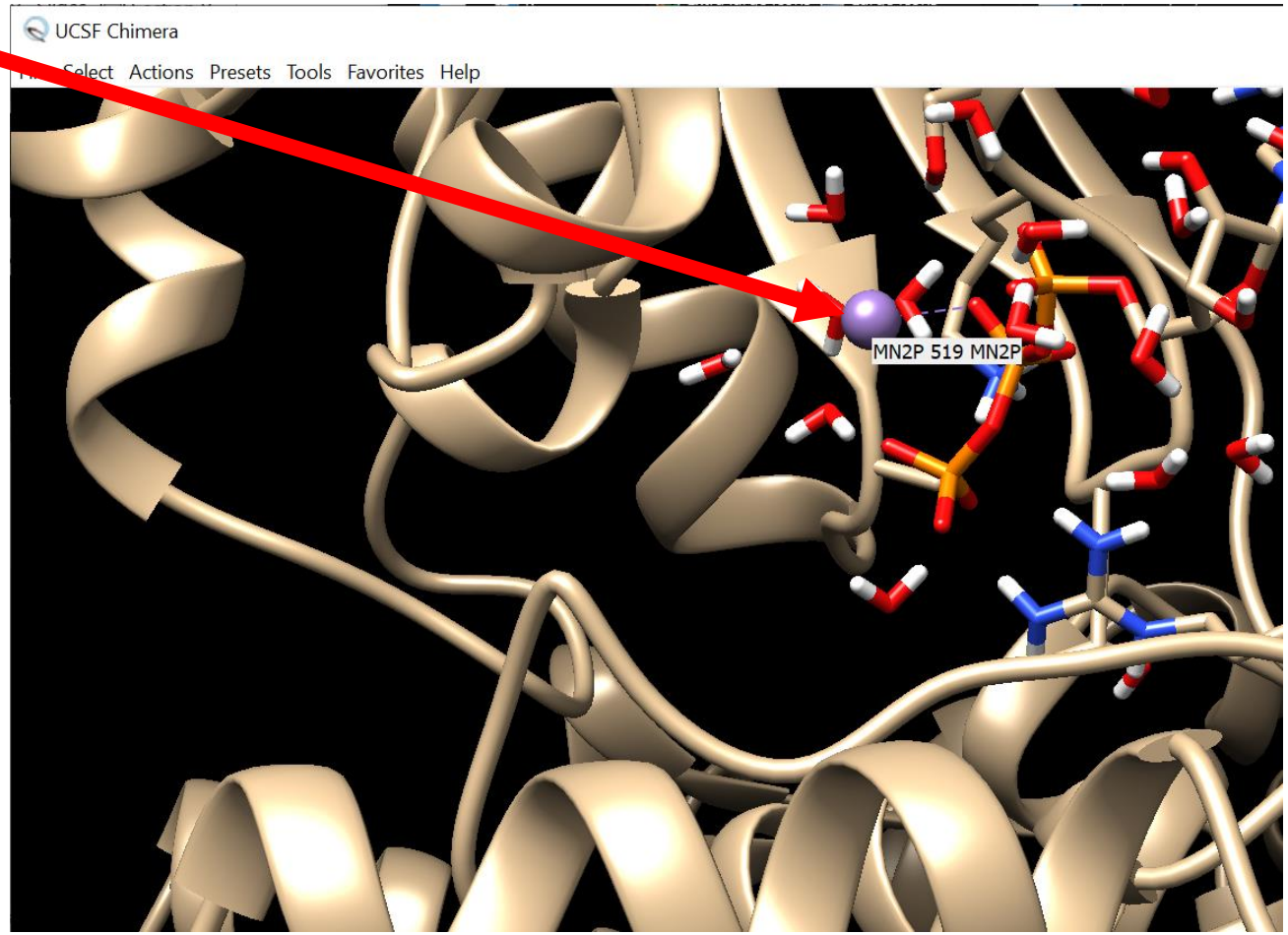
- Open the gromacs folder after extracting the files from CHARMM-GUI
- Open the folder with PowerShell or cmd, execute:

\YOUR\PATH\TO\GROMACS\bin\gmx.exe make_ndx -f .\step3_charmm2gmx.pdb -o nrg_grp.ndx



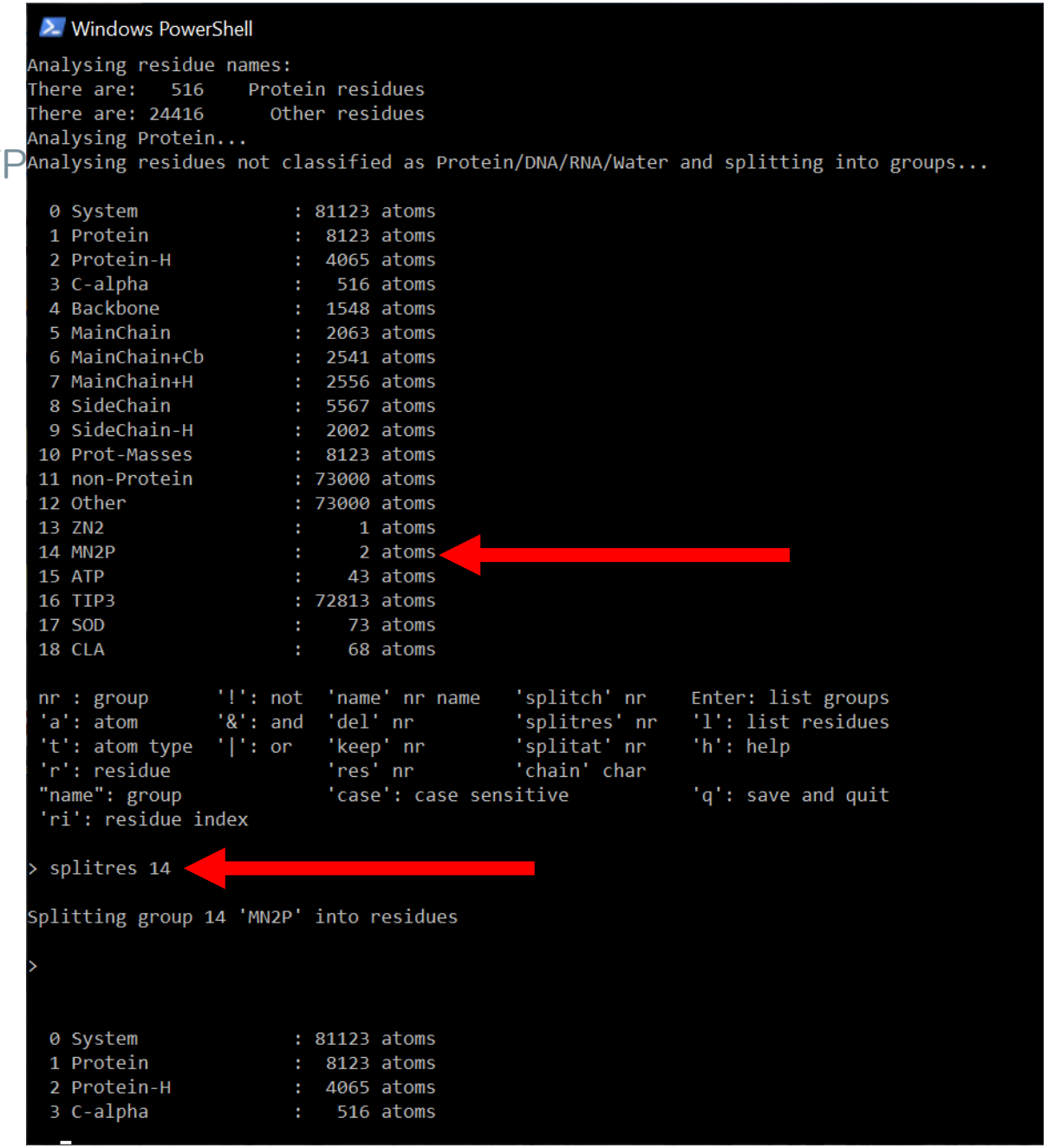
3bi. Generate index groups

- Open step3_charmm2gmx.pdb in Chimera
  - Find the residue number of this Mn



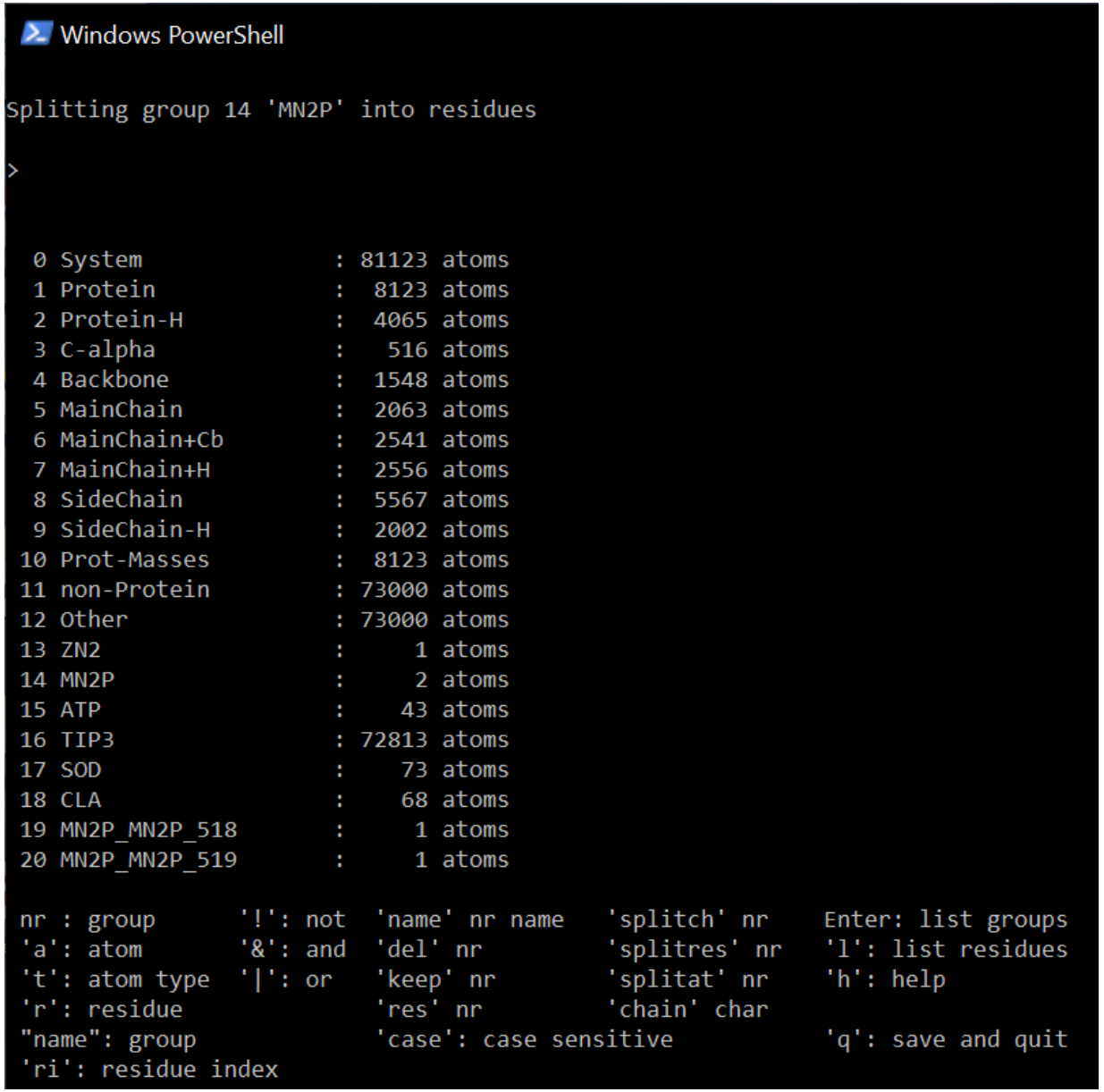
3bi. Generate index groups

- **Back to the command line**
  - Since the Mn^{2+} near ATP is a cofactor for ATP hydrolysis instead of a part of receptor (see DOI: 10.1093/emboj/cdg500), we need to separate it before generating the index groups.



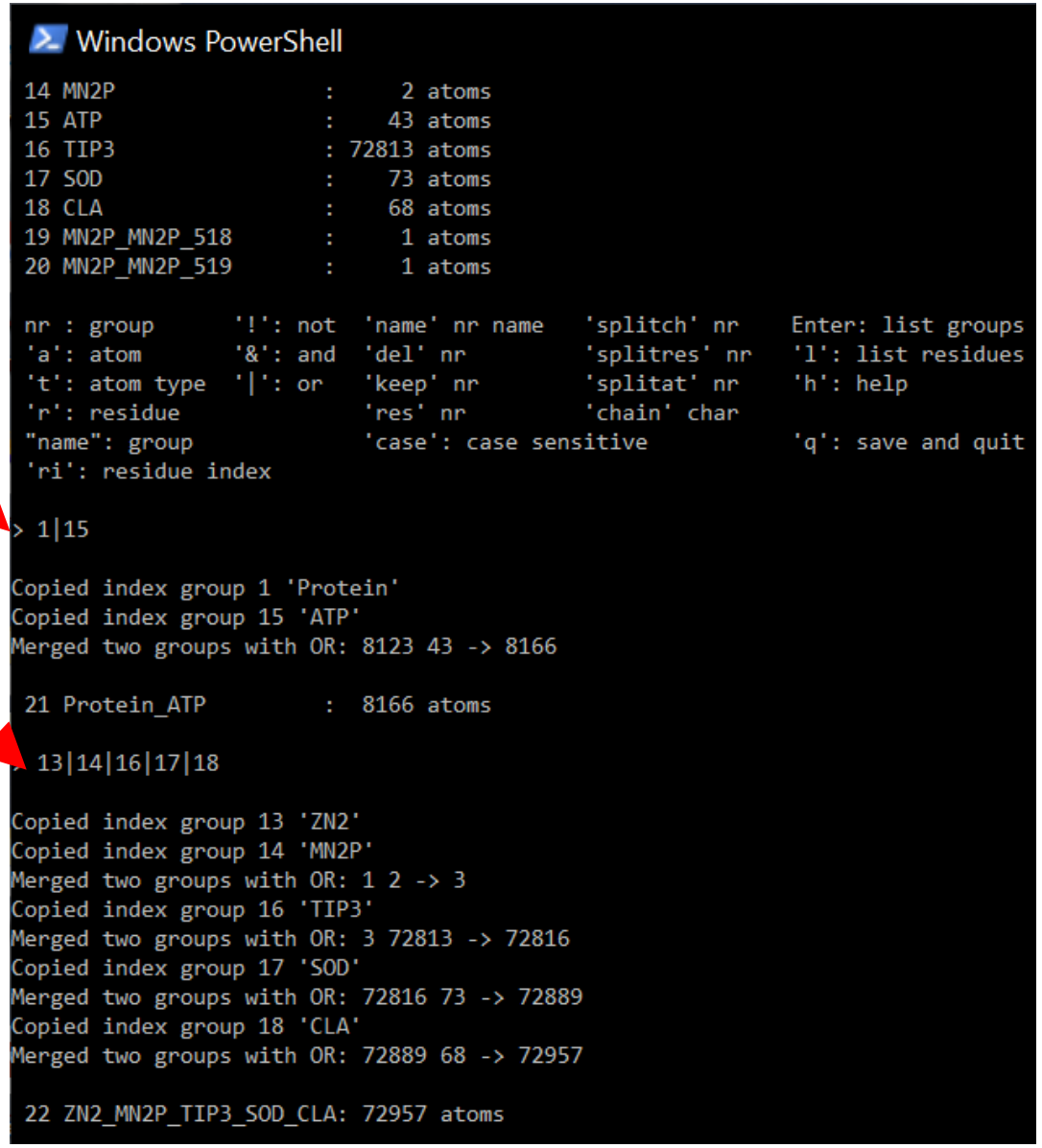
3bi. Generate index groups

- Press enter to see the updated list
- Now we can start making the index groups.



3bi. Generate index groups

- First we need to get back **the temperature-coupling groups in the index.ndx** generated by charmm-gui. This includes:
  - the group of protein + substrate
  - the group of solvent + ions



3bi. Generate index groups

- Then we will make index groups for interaction energy calculation.
  - Receptor: protein + Zn + Mn 518
  - Ligand: ATP

```
> 1|13|19

Copied index group 1 'Protein'
Copied index group 13 'ZN2'
Merged two groups with OR: 8123 1 -> 8124
Copied index group 19 'MN2P_MN2P_518'
Merged two groups with OR: 8124 1 -> 8125

 23 Protein_ZN2_MN2P_MN2P_518:  8125 atoms

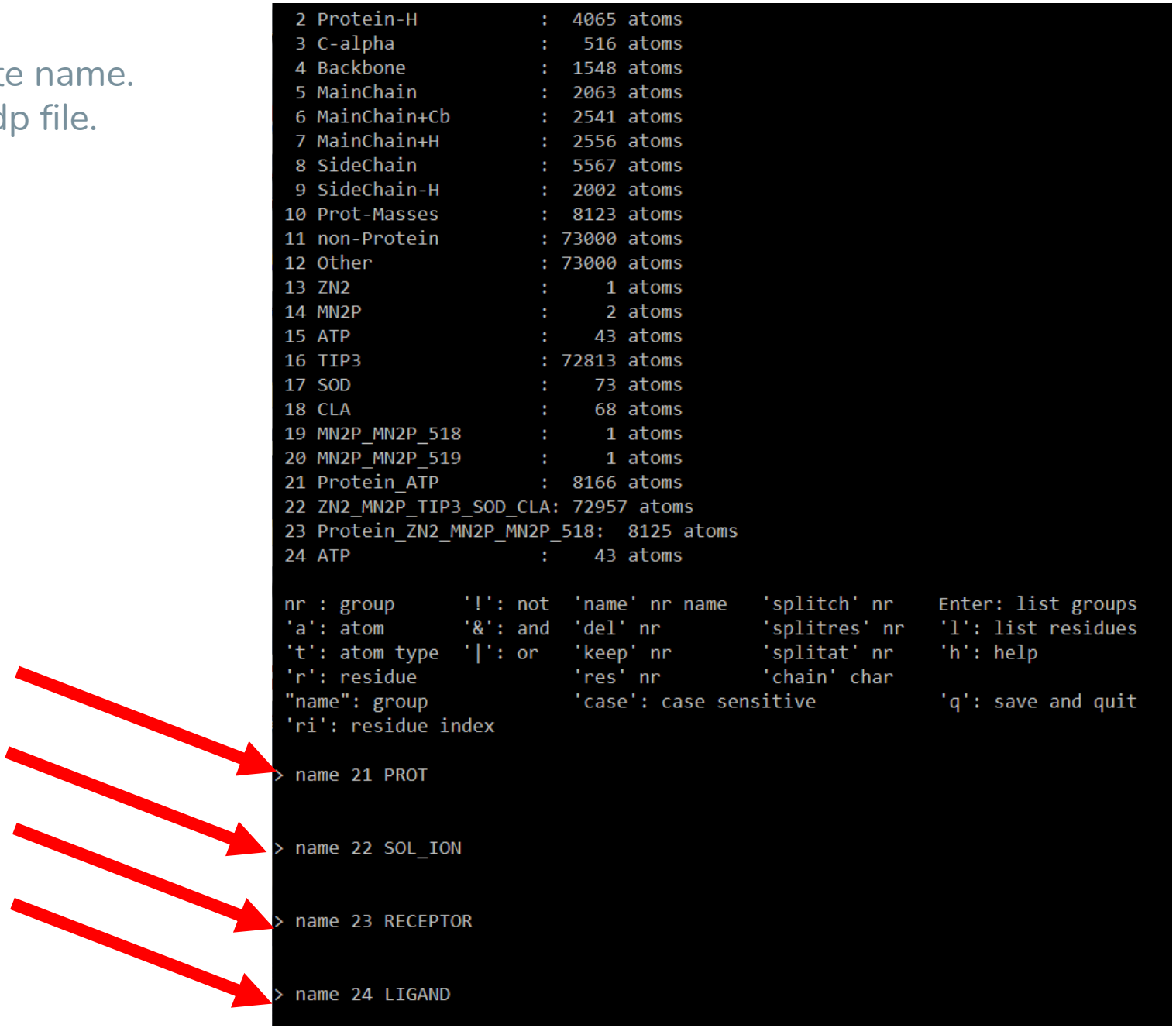
> 15

Copied index group 15 'ATP'

 24 ATP                 :    43 atoms
```

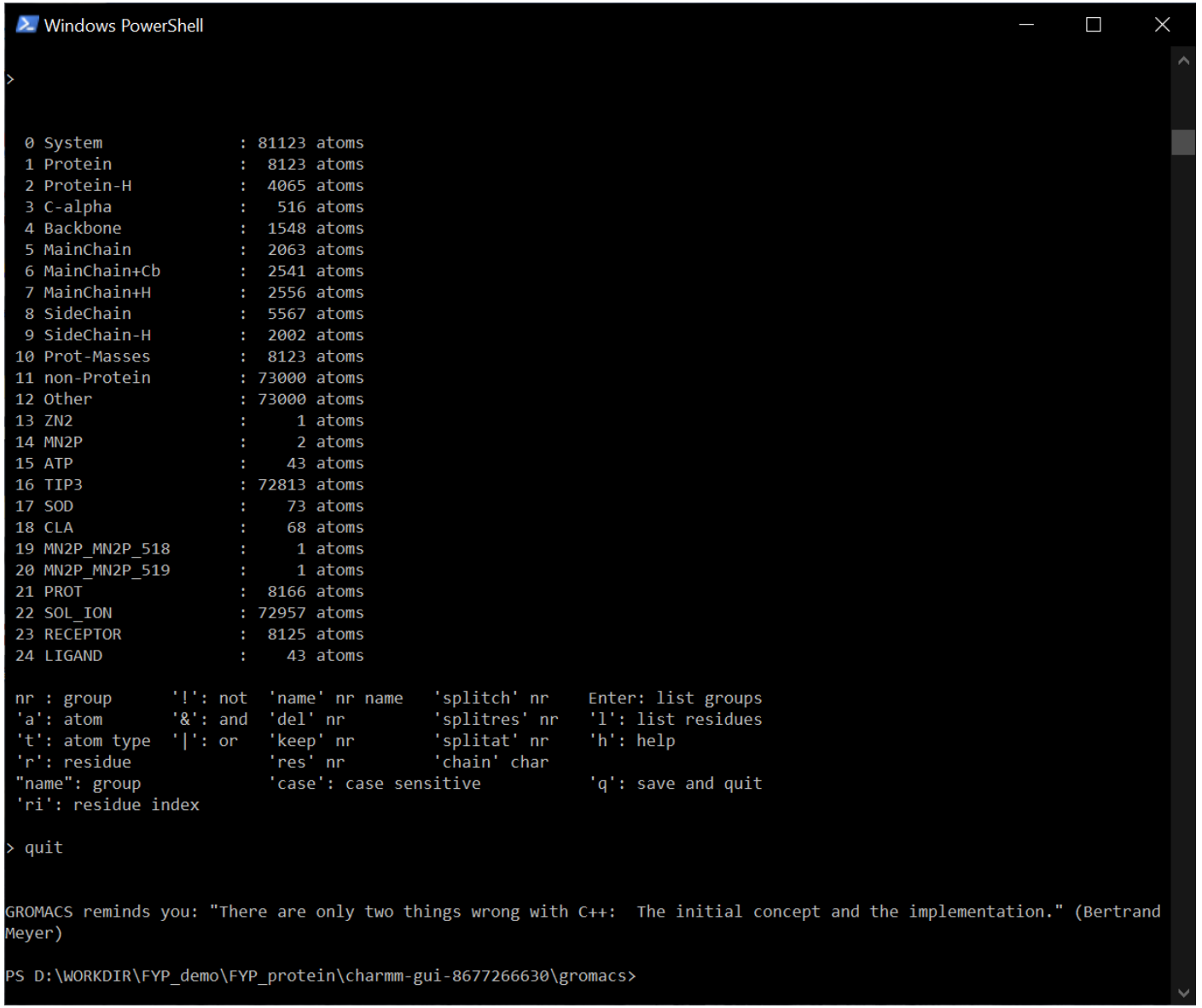
3bi. Generate index groups

- Give our new groups an appropriate name. We will use these names in our mdp file.



3bi. Generate index groups

- Press enter to check the updated list
- Now we are good to go! Type quit to generate the index file.
- But do not close the windows powershell yet!



3c. Add a mdp file with energy groups for interaction energy calculation

- Copy step5_production.mdp to step6_ie.mdp
- Add the following line to step6_ie.mdp:

```
energygrps                  = RECEPTOR LIGAND
```

6. Analysis

- Now you have your step6_ie.mdp ready.
- Execute the following commands one by one in the windows powershell (replace %GMXPATH% with your path to gmx.exe

```
%GMXPATH%\gmx.exe grompp -f step6_ie.mdp -c step4.2_equilibration.gro -p topol.top -n nrg_grp.ndx -o step6_ie.tpr
%GMXPATH%\gmx.exe mdrun -deffnm step6_ie -rerun step5_md.xtc -nb cpu
ECHO Coul-SR:RECEPTOR-LIGAND | %GMXPATH%\gmx.exe energy -f step6_ie.edr -o Coul-SR.xvg
ECHO LJ-SR:RECEPTOR-LIGAND | %GMXPATH%\gmx.exe energy -f step6_ie.edr -o LJ-SR.xvg
```

6. Analysis

- After the simulation ends (maybe half a day?):
  - Scroll up a bit, write down the average LJ-SR and Coul-SR energy with error estimation
  - Add them together to get the total interaction energy

Energy	Average	Err.Est.	RMSD	Tot-Drift	
LJ-SR:RECEPTOR-LIGAND	?	?	?	?	(kJ/mol)

Energy	Average	Err.Est.	RMSD	Tot-Drift	
Coul-SR:RECEPTOR-LIGAND	?	?	?	?	(kJ/mol)

- **Note that this is NOT the free energy or binding energy that are generally used -** just the decomposed short-range interaction energy that may or may not be useful, depends on force fields.
  - Usually more complicated methods are used for a more accurate analysis.
  - However it is quick and maybe useful in your case.